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Imperial College London

Department of Mathematics

Second-year Group Research Project

Cylindrical Wiener Process and Stochastic Heat Equation

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June 13, 2026

Abstract

A Stochastic Differential Equation is a type of differential equation that is perturbed by randomness, usually introduced through what is called the Wiener process (or Brownian Motion). They are commonly used to model physical systems in thermodynamics or dynamics of asset prices in mathematical finance. Although they work with the standard measure-theoretic setup in finite dimensional spaces, they cannot extend naturally to infinite dimensional spaces due to the limitation of translation invariance. This project explores the use of the cylindrical Wiener process to resolve this limitation and applies it to the stochastic heat equation. We then simulate, through numerical methods, the spectral form of the equation and investigate how the initial condition and the covariance structure affect the distribution of the solution.

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1 Introduction

Many physical systems are naturally modelled by partial differential equations, and many of these systems are subject to random disturbances. The stochastic analysis developed for ordinary differential equations, in which noise is supplied by a finite-dimensional Brownian motion [1], does not carry over directly to this setting: the state of a partial differential equation is a function rather than a vector, so the noise must live in an infinite-dimensional function space, and it is a classical fact that the most natural candidate, white noise with identity covariance, does not exist as a genuine random variable on such a space. The standard resolution, developed in the theory of stochastic equations in infinite dimensions [2, 3], is to weaken the notion of a measure to that of a *cylindrical measure*, which retains only the finite-dimensional projections of the would-be distribution, and correspondingly to replace the driving Brownian motion by a *cylindrical Wiener process* [4].

This project gives an exposition of these ideas, and applies it to a concrete example: the stochastic heat equation on the interval $[0, 1]$. The report is structured in two halves. The first half develops the theory. ?? motivates the problem through the heat equation; ?? recall the finite-dimensional theory of Wiener processes, Itô integration and stochastic differential equations that the infinite-dimensional theory imitates; and ?? explain why this theory breaks down in infinite dimensions, there is no useful infinite-dimensional measure, and the standard Gaussian fails to be countably additive, and build the cylindrical framework that repairs it, following [3–5]. The second half puts the theory into practise. Writing the stochastic heat equation as a stochastic evolution equation driven by a cylindrical Wiener process, we obtain its mild solution as a system of independent Ornstein–Uhlenbeck processes in the Fourier modes, and use this diagonal structure to simulate the equation by a spectral method. We investigate numerically how the initial condition and the covariance operator each shape the solution, characterise the distribution of the solution through Monte Carlo estimation of its mean and variance, and compare an Euler–Maruyama-type approximation of the noise increments against exact sampling from the Ornstein–Uhlenbeck transition distribution.

2 Motivation

We are interested in partial differential equations that include a “noise” term, and we take the heat equation as our main example. Before such an equation can be written down, some care is needed about what exactly our noise is. In our case the difficulty is that our “noise” is a function changing through time rather than a single number, so the usual notion of noise must be adapted to this setting. This introduction explains the need for the setup that makes this infinite-dimensional noise valid.

Consider heat conduction along a thin insulated rod of length one. Writing $u(t, x)$ for the temperature at position $x \in [0, 1]$ and time t , the temperature evolves according to the heat equation

$$\partial_t u = \partial_{xx} u = \Delta u$$

with the ends held at fixed temperature and an initial profile $u(0, \cdot) = u_0$. This equation is deterministic: for any initial set of conditions, the temperature at every later time is determined, and any irregularity in u_0 decays as the rod settles towards equilibrium as time goes to infinity.

In many situations this deterministic description is too idealised. A rod in contact with a fluctuating environment, or subject to a heat source that varies randomly in time and position, has fluctuations in heat throughout its length. To model this we would like to add a noise term to the equation and then ask whether there is a solution—or rather a distribution of solutions—and,

if so, what it looks like. To decide what this noise term should be, it is helpful to first recall how noise is introduced in finite dimensions.

In the finite-dimensional theory, randomness is added to an ordinary differential equation through a stochastic differential equation of the form

$$dX_t = f(t, X_t) dt + g(t, X_t) dB_t$$

The term $f(t, X_t) dt$ is the deterministic part, which carries the state along, and the term $g(t, X_t) dB_t$ is the random part, which disturbs it. The state X_t takes values in $\mathbb{R}^d$, and the noise is supplied by a d -dimensional Brownian motion $B_t = (B_t^{(1)}, \dots, B_t^{(d)})$, that is, by d independent scalar Brownian motions, one in each coordinate direction. The point to retain is that the noise takes values in the same space $\mathbb{R}^d$ as the state: in each direction of the state space there is an independent source of noise.

We now return to the heat equation and apply the same principle. At a fixed time t , the state of the system is not a point of $\mathbb{R}^d$ but the entire temperature profile $u(t, \cdot)$, which is a function on $[0, 1]$ and hence an element of an infinite-dimensional space, such as $L^2([0, 1])$. If, as in finite dimensions, the noise is to take values in the same space as the state, then it too must be infinite-dimensional, with an independent scalar Brownian motion in each “direction”. The difference is that the space now has infinitely many directions, so we are led to consider infinitely many independent Brownian motions at once.

This is the object the noise term has to encode, and it is exactly here that the finite-dimensional recipe stops working. The natural thing to try is to collect the Brownian motions into a single process living in $L^2([0, 1])$, the way the scalar Brownian motions in $\mathbb{R}^d$ are simply stacked into B_t . Concretely, one fixes an orthonormal basis $\{e_k\}$ of $L^2([0, 1])$ and forms $\sum_k B_t^{(k)} e_k$ from independent scalar Brownian motions. In infinite dimensions this sum does not converge in $L^2([0, 1])$: at a fixed time each term contributes the same amount of variance, and with infinitely many directions the total expected squared length is infinite. The same fact can be read off from the covariance. A Gaussian on a Hilbert space is determined by its covariance matrix, and for an honest Gaussian measure that operator must be trace-class (the diagonal sum must be finite). The object we are after has covariance equal to the identity, whose trace is the divergent sum $\sum_k 1$, so no genuine Gaussian measure has it as its covariance. This is not something a cleverer choice of space can fix: the constructions behind Brownian motion in $\mathbb{R}^d$ do not extend to infinite-dimensional spaces.

These difficulties motivate the framework developed below. Instead of requiring a genuine measure, or a process taking values in $L^2([0, 1])$, one works with cylindrical measures (the finite-dimensional projections of our process), which retain exactly the finite-dimensional information that is well defined and nothing more. The following sections set up Wiener processes and cylindrical measures, and then return to defining infinite-dimensional stochastic partial differential equations such as the stochastic heat equation.

3 Setup

3.1 Classical Brownian Motion

For the purposes of the stochastic perturbation of differential equations which are the focus of this study, we require a way to model white noise in a standard way.

Definition 3.1. A Gaussian process is a stochastic process $\{\xi_t : t \in T\}$ where all finite collections $\{\xi_{t_1}, \xi_{t_2}, \dots, \xi_{t_n}\}$ of indices of T are multivariate Gaussian distributions.

We build upon this definition by imbuing it with extra properties that derive a process that acts like the noise we wish to simulate.

Definition 3.2. A Gaussian process $\{W(t) : t \in \mathbb{R}_{\geq 0}\}$ of random variables $W(t)$ is called a *Wiener process over $\mathbb{R}$* if:

W0: $W(0) = 0$ almost surely (a.s.)

W1 (Independent increments): For $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $W(t_1) - W(t_0), \dots, W(t_n) - W(t_{n-1})$ are independent

W2 (Stationary increments): For $s < t$, $W(t) - W(s) \sim \mathcal{N}(0, t - s)$

W3: $[0, \infty) \ni t \mapsto W(t) \in \mathbb{R}$ is continuous a.s.

Equivalently, a real-valued Wiener process is a centered Gaussian process with covariance

$$\text{Cov}(W(s), W(t)) = \mathbb{E}[W(s)W(t)] = \min(s, t), \quad s, t \geq 0$$

This covariance point of view will provide a useful way to extend naturally to higher-dimensional and infinite-dimensional settings.

These properties provide a way to generate noise during simulations by polling normal distributions with variance given by the simulation's time increment. However, the current definition in a similar way to measures is not powerful enough to describe processes over infinite dimensions. For this case we require a general analogue, which we can adapt into the cylindrical case.

4 Finite-Dimensional Stochastic Differential Equations

Before turning to noise in infinite dimensions, we recall the finite-dimensional theory that motivates it. A stochastic differential equation describes a system whose evolution has both a deterministic trend and a random perturbation: the drift term sets the direction in which the state is carried, while the diffusion term injects noise driven by a Wiener process. Solutions are understood through the Itô integral against this process, and in general cannot be written in closed form, so they are studied through numerical approximation. We set out these ingredients here in the scalar case before adapting them to function spaces.

4.1 Classical Itô Stochastic Differential Equations

4.1.1 Finite-Dimensional Numerical Approximation

Classical Itô stochastic differential equations take the form

$$dX(t) = f(t, X(t))dt + g(t, X(t))dW(t)$$

where the state $X(t)$ takes values in $\mathbb{R}^d$ and the driving noise $W(t)$ is a real-valued Wiener process [1].

4.1.2 Itô Integration

Definition 4.1. When finding solutions or estimations to stochastic differential equations as above, we require a notion of integration over a Wiener process. These integrals take the form

$$\int_a^b g(t, X(t)) dW(t)$$

and are called *Itô integrals*. They are defined as a limit of integrals of elementary integrands. For g that is piecewise constant in time, $g(t, X(t)) = \sum_i g_{t_i} \mathbf{1}_{(t_i, t_{i+1}]}(t)$ with each g_{t_i} depending only on time steps up to t_i , the integral is the sum

$$\int_a^b g(t, X(t)) dW(t) = \sum_i g_{t_i} (W(t_{i+1}) - W(t_i)),$$

which satisfies the *Itô isometry*

$$\mathbb{E} \left[\left(\int_a^b g(t, X(t)) dW(t) \right)^2 \right] = \mathbb{E} \left[\int_a^b g(t, X(t))^2 dt \right].$$

A general integrand g with $\mathbb{E} \left[\int_a^b g(t, X(t))^2 dt \right] < \infty$ is approximated by such elementary integrands $g^{(n)}$ in this norm, and the isometry ensures the integral is well defined as the limit

$$\int_a^b g(t, X(t)) dW(t) = \lim_{n \rightarrow \infty} \int_a^b g^{(n)}(t, X(t)) dW(t)$$

and can be estimated with the sequence

$$I_{t_n} = I_{t_{n-1}} + g(t_{n-1}, X(t_{n-1})) W(\Delta t)$$

5 Candidate Function Spaces

Before choosing a state space in which the process lives, it is natural to first consider the spaces in which its sample paths live. Recall that a one-dimensional Wiener process $W(t)$ can be viewed as a random function of time. On a finite interval such as $[0, 1]$, it has some nice properties mentioned in [Definition 3.2](#), which indicate that its sample paths live in the following spaces:

- By **W3**, $W(t)$ is continuous almost surely, so one can regard a sample path as an element of $C[0, 1]$, the space of real continuous functions on $[0, 1]$;
- Moreover, since $W(0) = 0$ almost surely by **W0**, one can more precisely regard it as an element of $C_0[0, 1]$, the subspace of $C[0, 1]$ consisting of functions satisfying $f(0) = 0$;
- Since continuous functions on $[0, 1]$ are square-integrable, a sample path may also be regarded as an element of $L^2([0, 1])$.

As suggested in [\[3\]](#), when extending finite-dimensional Wiener processes to infinite-dimensional spaces, the state at each time is itself a function. Thus, the state space is no longer a finite-dimensional vector space, but a function space.

A commonly used class of spaces is the class of separable Banach spaces. To allow random states, we need the concept of a σ -algebra, and once a Banach space is equipped with a norm, it is natural to consider its Borel σ -algebra. In practice, it is desirable to assume separability, since

then the Borel σ -algebra is generated by open balls. This makes measure-theoretic arguments easier to handle.

The following are some examples:

- $C[0, 1]$, the space of real continuous functions on $[0, 1]$ with the sup norm

$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$$

- $C_0[0, 1]$, the subspace of $C[0, 1]$ consisting of functions satisfying $f(0) = 0$, also with the sup norm
- Separable Hilbert spaces, for example $L^2(\mathbb{R})$ and $\ell^2(\mathbb{R})$, where $\ell^2(\mathbb{R})$ is the space of all infinite real sequences with finite sum of squares

It is also possible to consider $\mathbb{R}^\infty$, the space of all real infinite sequences, but $\mathbb{R}^\infty$ has a weak topological structure. For example, the product topology on $\mathbb{R}^\infty$ is not normable, which means that it lacks a global metric with which to measure distances between random states.

6 The Obstruction to an Infinite-Dimensional Measure

Although separable Banach spaces have nice topological properties, it is very difficult to define measures in an infinite-dimensional Banach space due to the lack of local compactness. Specifically, we have the following result from [6, 7]:

Theorem 6.1. *Let U be an infinite-dimensional separable Banach space. There is no translation-invariant Borel measure on U that assigns positive finite measure to open balls. In fact, any translation-invariant measure in U is either the zero measure or assigns infinite measures to open balls.*

Proof. Suppose $(U, \|\cdot\|)$ is an infinite-dimensional separable Banach space. We use the notation

$$d(x, E) = \inf_{u \in E} \|x - u\|$$

where $x \in U$ and $E \subset U$ is a proper subset of U . For any open ball $B(x, r)$, we show that one can find infinitely many disjoint open balls with small fixed radius using similar arguments of Riesz's lemma.

Let $x_1 \in B(x, r)$ be arbitrary with $\|x_1 - x\| = r/2$. Suppose one has chosen $\{x_1, \dots, x_n\} \subset B(x, r)$. Define $U_n := \text{span}\{x_1 - x, \dots, x_n - x\}$. Since U is infinite-dimensional, U_n is a finite-dimensional closed proper subset of U . We choose $y \in U \setminus U_n$ so that $d(y, U_n) > 0$. By definition, there exists $z \in U_n$ such that $\|y - z\| < 2d(y, U_n)$. Choose $x_{n+1} := x + \frac{r}{2} \frac{y-z}{\|y-z\|}$, then $\|x_{n+1} - x\| = r/2$ and

$$\begin{aligned} d(x_{n+1} - x, U_n) &= \inf_{u \in U_n} \left\| u - \frac{r}{2} \frac{y-z}{\|y-z\|} \right\| \\ &= \frac{r}{2} \inf_{u \in U_n} \left\| \frac{2}{r} u - \frac{y-z}{\|y-z\|} \right\| \\ &= \frac{r}{2} \inf_{v \in U_n} \frac{\|v - y\|}{\|y-z\|} \quad \text{by letting } v = \frac{\|y-z\|}{r/2} u + z \in U_n \\ &= \frac{r}{2} \frac{d(y, U_n)}{\|y-z\|} \\ &> \frac{r}{4} \end{aligned}$$

Since $x_i - x \in U_n$ for all $i \leq n$, this implies that

$$\|x_{n+1} - x_i\| > \frac{r}{4} \quad \text{for all } i \leq n$$

By continuing the induction, we produce a sequence $\{x_i\}_{i \geq 1}$ with $\|x_i - x\| = r/2$ such that $\|x_i - x_j\| > r/4$ for $i \neq j$, and thus the balls $B(x_i, r/8)$ are pairwise disjoint. Moreover, all of these balls are all contained in $B(x, r)$.

Now let μ be a translation-invariant measure on U . By translation invariance, all the balls $B(x_i, r/8)$ have the same measure. If that measure is positive, then $\mu(B(x, r)) = \infty$, contradiction. If that measure is zero, then by separability U can be covered by countably many balls of radius $r/8$, so μ is the zero measure. Therefore there is no translation-invariant measure on U assigning positive finite measure to open balls. $\square$

The consequence of this theorem is that, unlike in finite-dimensional Euclidean spaces, one cannot define a Gaussian measure in infinite dimensions by a translation-invariant measure and its density. We shall see later that this barrier inspires a different projection-based approach to Gaussian measure and the notion of cylindrical measures.

7 Cylindrical Sets and Finite-Dimensional Projections

As mentioned previously, we will introduce the notion of cylindrical objects to accommodate the lack of useful measure in infinite dimensional Banach spaces. Many concepts from probability theory have an analogue in cylindrical form. For consistency, we will use similar definitions and notation to these articles [3–5]. Let U be a separable Banach space and denote its continuous dual space as U^* . Let $\Gamma \subseteq U^*$.

Definition 7.1. For a finite subset of continuous (equivalently, bounded) linear functionals $\{u_1^*, \dots, u_n^*\} \subseteq \Gamma$ and a Borel set $B \in \mathcal{B}(\mathbb{R}^n)$, the *cylindrical set* with base B and generator $\{u_1^*, \dots, u_n^*\}$ is

$$C(u_1^*, \dots, u_n^*, B) := \{u \in U : (u_1^*(u), \dots, u_n^*(u)) \in B\}$$

The family of sets with this form is called *cylindrical sets with respect to (U, Γ)* and forms an algebra. The generated σ -algebra is denoted as $\mathcal{C}(U, \Gamma)$ and is called the *cylindrical σ -algebra with respect to (U, Γ)* . We also write $\mathcal{C}(U) := \mathcal{C}(U, U^*)$, which is exactly equal to $\mathcal{B}(U)$, the Borel σ -algebra of U since U is separable.

Example 7.2. In $\mathbb{R}^3$ space, the unit cylinder $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 < 1\}$ is the cylindrical set with the open unit disk in $\mathbb{R}^2$ as the base and $\{e_1^*, e_2^*\}$ as the generator, where $e_1^*(x, y, z) = x$ and $e_2^*(x, y, z) = y$.

Informally, continuous linear functionals may be interpreted as generalised coordinate or “directional” measurements on U . Cylindrical sets are restricted on finitely many “directions” while all remaining “directions” are left unrestricted.

In Hilbert space H with inner product $\langle \cdot, \cdot \rangle$ and its dual H^* , the Riesz representation theorem states that any continuous linear functionals $u^* \in H^*$ satisfy $\forall x \in H, u^*(x) = \langle x, v \rangle$ for a unique vector $v \in H$. Thus, cylindrical sets correspond to restrictions on finitely many “directions”.

8 Cylindrical Measures and Cylindrical Random Variables

Definition 8.1. A cylindrical measure on $\mathcal{C}(U)$ is a function $\mu : \mathcal{C}(U) \rightarrow [0, \infty]$ that has its restriction to the σ -algebra $\mathcal{C}(U, \Gamma)$ being a measure for any finite $\Gamma \subseteq U^*$.

In general, μ requires only finite additivity but not necessarily countable additivity, so it may not be a measure. It behaves like a measure when restricted to a finite-dimensional cylindrical σ -algebra. Similarly to any measure, μ is called finite if $\mu(U) < \infty$ and μ is a cylindrical probability measure if $\mu(U) = 1$.

If a function $f : U \rightarrow \mathbb{R}$ is measurable with respect to $\mathcal{C}(U, \Gamma)$ for some finite $\Gamma \subseteq U^*$, then f depends only on finitely many coordinates. In that case, integration with respect to μ can be realised through the corresponding finite-dimensional image measure, whenever the integral exists. Thus, the characteristic function of a finite cylindrical measure is well-defined.

Definition 8.2. The characteristic function (or Fourier Transform) $\varphi_\mu : U^* \rightarrow \mathbb{C}$ of a finite cylindrical measure μ is

$$\varphi_\mu(u^*) := \int_U e^{iu^*(u)} d\mu(u) \quad \text{for all } u^* \in U^*$$

Definition 8.3. For a finite set $\Gamma = \{u_1^*, \dots, u_n^*\} \subseteq U^*$ and a cylindrical measure μ ,

$$\mu_\Gamma : \mathcal{B}(\mathbb{R}^n) \rightarrow [0, \infty], \quad \mu_\Gamma(B) := \mu(C(u_1^*, \dots, u_n^*, B))$$

is called the image measure of μ under the mapping $u \mapsto (u_1^*(u), \dots, u_n^*(u))$. Since μ is restricted to $\mathcal{C}(U, \Gamma)$, μ_Γ defines a measure on $\mathcal{B}(\mathbb{R}^n)$ by [Theorem 8.1](#).

The characteristic function $\varphi_{\mu_\Gamma} : \mathbb{R}^n \rightarrow \mathbb{C}$ of μ_Γ satisfies

$$\begin{aligned} \varphi_{\mu_\Gamma}(\beta_1, \dots, \beta_n) &= \int_{\mathbb{R}^n} e^{i(\beta_1 x_1 + \dots + \beta_n x_n)} d\mu_\Gamma(x_1, \dots, x_n) \\ &= \int_U e^{i(\beta_1 u_1^*(u) + \dots + \beta_n u_n^*(u))} d\mu(u) \\ &= \varphi_\mu(\beta_1 u_1^* + \dots + \beta_n u_n^*) \end{aligned}$$

for all $\beta_1, \dots, \beta_n \in \mathbb{R}$.

Similarly to measures, cylindrical measures can be uniquely described by their characteristic functions. Equivalently, they can be uniquely described by the family of one-dimensional image measures $\{\mu_{u^*} : u^* \in U^*\}$, since their characteristic functions satisfy

$$\varphi_\mu(u^*) = \int_{\mathbb{R}} e^{ix} d\mu_{u^*}(x) \quad \text{for all } u^* \in U^*$$

Theorem 8.4 (Bochner-Kolmogorov's Theorem). A function $\varphi : U^* \rightarrow \mathbb{C}$ is a characteristic function of some cylindrical measure in U if and only if

1. $\varphi(0) = 1$;

2. (positive-definiteness) For any $n \in \mathbb{N}$, any $\xi_1, \dots, \xi_n \in \mathbb{C}$ and any $u_1^*, \dots, u_n^* \in U^*$,

$$\sum_{i=1}^n \sum_{j=1}^n \varphi(u_i^* - u_j^*) \xi_i \bar{\xi}_j \geq 0;$$

3. (pseudo-continuity) φ is continuous on every finite-dimensional subspace of U^* .

Proof. See [8, IV.4.2] □

Definition 8.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space. A cylindrical random variable $X : U^* \rightarrow L^0(\Omega, \mathcal{F}, P; \mathbb{R})$ is a continuous linear map that maps continuous linear functionals to random variables in $(\Omega, \mathcal{F}, P)$. Its corresponding characteristic function $\varphi_X : U^* \rightarrow \mathbb{C}$ is

$$\varphi_X(u^*) := \mathbb{E} \left[e^{iX(u^*)} \right] \quad \text{for all } u^* \in U^*$$

Naturally, the cylindrical distribution $\mu_X : \mathcal{C}(U) \rightarrow [0, 1]$ of X is

$$\mu_X(C(u_1^*, \dots, u_n^*, B)) = P((X(u_1^*), \dots, X(u_n^*)) \in B)$$

Since genuine U -valued random variables may fail to exist, cylindrical random variables are defined to behave like random variables only when observed through finitely many continuous linear functionals.

Definition 8.6. A cylindrical process X in U is a family $\{X(t) : t \in \mathbb{R}_{\geq 0}\}$ of cylindrical random variables in U .

8.1 Gaussian Cylindrical Measures

As mentioned previously, cylindrical measures can be uniquely defined from the set of one-dimensional image measures. To allow extensions to stochastic process and maintain consistency among finite restrictions, it is natural to have these image measures following the same distribution.

To obtain a coherent cylindrical distribution from one-dimensional projections, one needs compatibility under linear combinations of functionals. Gaussian cylindrical measures meet the requirement because Gaussian distributions are preserved under linear combinations, making them natural candidates for consistent finite-dimensional projections. [9]

Definition 8.7. A cylindrical measure μ on U is called a *weakly Gaussian cylindrical measure* if μ_{u^*} is a Gaussian measure on $\mathcal{B}(\mathbb{R})$ for all $u^* \in U^*$. This is equivalent to μ_Γ being a Gaussian measure on $\mathcal{B}(\mathbb{R}^n)$ for all finite $\Gamma = \{u_1^*, \dots, u_n^*\} \subseteq U^*$ by the Cramér-Wold theorem.

Theorem 8.8 (4, Theorem 3.2). The characteristic function of a weakly Gaussian cylindrical measure $\varphi_\mu : U^* \rightarrow \mathbb{C}$ is given by

$$\varphi_\mu(u^*) = \int_{\mathbb{R}} e^{ix} d\mu_{u^*}(x) = \exp \left(im(u^*) - \frac{1}{2}s(u^*) \right)$$

for all $u^* \in U^*$, where $m : U^* \rightarrow \mathbb{R}$ is a linear functional representing the mean, and $s : U^* \rightarrow \mathbb{R}_{\geq 0}$ is a quadratic form representing the variance.

If there exists a bounded and positive symmetric linear operator $Q : U^* \rightarrow U$ such that the quadratic form can be written as $s(u^*) = \langle Qu^*, u^* \rangle$, then Q is called the *covariance operator* (analogue to covariance matrix), and μ is referred to as a *strongly Gaussian cylindrical measure*.

In a Hilbert space, one can define the standard Gaussian cylindrical measure through its characteristic function.

Definition 8.9. Let H be a separable Hilbert space with norm $\|\cdot\|$, the *standard Gaussian cylindrical measure* γ on H has the characteristic function

$$\varphi_\gamma : H \rightarrow \mathbb{C}, \quad \varphi_\gamma(u) = e^{-\frac{1}{2}\|u\|^2}$$

Lemma 8.10. If H is infinite-dimensional, γ is not a measure.

Proof. See [6, Proposition 4.38] □

Remark 8.11. Formally, the characteristic function should be defined in the dual space H^* , but one can identify $H \cong H^*$ using the Riesz representation theorem to remove the distinction. In this case, you can apply [Theorem 8.8](#) and choose $m = 0$ and $s = \|\cdot\|$ to obtain the formulation of the dual space. Such an identification fails in general Banach space.

Another benefit of stable distribution is to allow the use of the Central Limit Theorem in infinite dimensional space under certain constraints. Recall that the Lindeberg-Lévy Central Limit Theorem states that for any sequence $X_1, \dots, X_n$ of i.i.d. random vectors in $\mathbb{R}^k$ with $\mathbb{E}[X_i] = \mu$ and a positive-definite $k \times k$ covariance matrix $\Sigma = \text{Var}[X_i]$, we have

$$\frac{X_1 + \dots + X_n - n\mu}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, \Sigma)$$

However, the classical Central Limit Theorem fails in infinite dimensional Banach space [10]. In Hilbert spaces, the inner-product structure provides stronger geometric and probabilistic tools. Under suitable moment conditions, one can use the Central Limit Theorem.

Theorem 8.12 (11, Section 3). Let H be a separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$ and X be a centered H -valued random variable. Let $X_1, X_2, \dots$ be sequence of i.i.d. copies of X such that $\mathbb{E}[\|X_i\|^2] < \infty$. Let $Q : H \rightarrow H$ be the covariance operator of X , then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \xrightarrow{d} \mathcal{N}(0, Q)$$

In this setting, Q is automatically positive, self-adjoint and **trace-class**, i.e.

$$\text{tr}(Q) = \sum_{j=1}^{\infty} \langle Qe_j, e_j \rangle < \infty$$

for some (hence any) orthonormal basis $\{e_j\}$.

9 Cylindrical Wiener Processes

With the theory of Cylindrical measures, we can now expand the definitions from [Subsection 3.1](#) to infinite dimensions.

9.1 Q -Wiener Processes and Cylindrical Wiener Processes

Definition 9.1. A process $W(t)$ over a Hilbert space H is called a Q -Wiener Process over H if it follows **W0**, **W1**, and **W3** over H as in the one dimensional case, and follows an updated

W2*: for $s < t$, $W(t) - W(s) \sim \mathcal{N}(0, (t - s)Q)$

where Q is the covariance operator defined such that

$$t\langle Qu, v \rangle = \mathbb{E}[\langle W(t), u \rangle \langle W(t), v \rangle]$$

This definition allows us to extend processes to infinite dimensions, but only given certain conditions for the covariance matrix Q : indeed, if Q is not trace-class, we cannot define a random variable in the standard way. The process we wish to analyse is a Q -Wiener process over an infinite dimensional function space with covariance $Q = I$, which does not fit into this category. In this scenario, we must move to the cylindrical case.

Definition 9.2. Let $W(t)$ be a Q -Wiener process over a Hilbert space H with inner product $\langle \cdot, \cdot \rangle$. For each $u \in H$, define a new Wiener process $W_u(t)$ over $\mathbb{R}$ defined as

$$W_u(t) = \langle u, W(t) \rangle$$

If $Q = I$, the linear transformation $u \rightarrow W_u$ is called a *cylindrical Wiener process*. Equivalently, it can be defined in terms of linear functional $u^* \in H^*$:

$$W_{u^*}(t) = u^*(W(t))$$

If we select a $t \in \mathbb{R}_{\geq 0}$, the function

$$u^* \mapsto u^*(W(t))$$

is a cylindrical random variable. That is, cylindrical Wiener processes are cylindrical processes [2].

As cylindrical Wiener processes operate via a covariance matrix $Q = I$, it follows naturally that they induce strongly Gaussian cylindrical measures.

Theorem 9.3. Let $W(t)$ be a cylindrical Wiener process. Then for all $t \in [0, \infty)$, the induced cylindrical measure $\mu_{W(t)}$ is strongly Gaussian.

Proof. It is clearly weakly Gaussian as

$$\mu_{W(t)}(C(u^*, B)) = P(u^*(W(t)) \in B)$$

and $u^*(W(t))$ is Gaussian by definition. Using **W2** in [Definition 9.1](#), we find that the mean $m = 0$ and the variance $s(u^*) = t\langle Qu^*, u^* \rangle$, which implies that the measure is also strongly Gaussian. $\square$

9.2 A Fourier Basis Example in $L^2([0, 1])$

Example 9.4. Let $H = L^2([0, 1])$ be the squared Lebesgue integrable functions over $[0, 1]$ and its dual H^* , and let $e_n = \sin(n\pi x)$ be its orthonormal basis via the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. Then if $u \in H$, then

$$u(x) = \sum_{n=0}^{\infty} \lambda_n \sin(n\pi x)$$

Let $u_n^*(u) = \lambda_n = \langle u, e_n \rangle \in H^*$ be the projection maps. Consider a H -valued cylindrical Wiener process $W(t)$. Then

$$\begin{aligned} \mu_{W(t)}(C(u_1^*, u_2^*, \dots, u_n^*, B)) &= P((u_1^*W(t), u_2^*W(t), \dots, u_n^*W(t)) \in B) \\ &= P((\langle W(t), e_1 \rangle, \langle W(t), e_2 \rangle, \dots, \langle W(t), e_n \rangle) \in B) \end{aligned}$$

As each $\langle W(t), e_n \rangle$ is a Gaussian random variable over $\mathbb{R}$, we can compute

$$\text{Cov}(\langle W(t), e_n \rangle, \langle W(t), e_m \rangle) = 0$$

for $n \neq m$ as the covariance matrix $Q = I$. So each $\langle W(t), e_n \rangle$ is independent, giving a covariance matrix $\Sigma_{ii} = t$, and thus

$$\mu_{W(t)}(C(u_1^*, u_2^*, \dots, u_n^*, B)) = \int_B \frac{1}{(2\pi)^{n/2} t^{n/2}} e^{-\frac{t}{2} \|x\|^2} dx$$

More generally, cylindrical Wiener processes are Gaussian over a finite set of basis vectors.

10 From Finite-Dimensional Noise to Infinite-Dimensional Noise

Now that we have the constructions allowing us to define an infinite dimensional random process with arbitrary covariance, we can generalize the theory from [Section 4](#).

10.1 Stochastic Evolution Equations on Hilbert Spaces

As discussed, for a Hilbert space H , a natural candidate would be an H -valued Wiener process with covariance operator $Q = I$, but this fails to exist as a genuine random variable when H is infinite-dimensional. The cylindrical Wiener process resolves this obstruction and serves as our driving noise for stochastic partial differential equations on H .

Definition 10.1. A family $\{S_t\}_{t \geq 0}$ of bounded linear operators on a separable Hilbert space H is called a *strongly continuous semigroup* (or C_0 -semigroup) if

1. $S_0 = I$, where I is the identity operator on H ;

2. $S_{s+t} = S_s S_t$ for all $s, t \geq 0$;
3. $\lim_{t \rightarrow 0^+} \|S_t x - x\| = 0$ for all $x \in H$.

The *infinitesimal generator* A of $\{S_t\}_{t \geq 0}$ is the linear operator with domain

$$D(A) := \left\{ x \in H : \lim_{t \rightarrow 0^+} \frac{S_t x - x}{t} \text{ exists} \right\}$$

defined by

$$Ax := \lim_{t \rightarrow 0^+} \frac{S_t x - x}{t} \quad \text{for all } x \in D(A).$$

By analogy with the scalar case, we write $S_t = e^{tA}$.

Definition 10.2. Let A be the infinitesimal generator of a C_0 -semigroup $\{S_t\}_{t \geq 0}$ on a separable Hilbert space H , let W be the cylindrical Wiener process on H , and let B be a bounded linear operator on H . A *stochastic evolution equation* on H is an equation of the form

$$dX(t) = AX(t)dt + BdW(t), \quad X(0) = x_0$$

whose solution $\{X(t)\}_{t \geq 0}$ is an H -valued process with given initial condition $x_0 \in H$.

More generally one may include a nonlinear drift $F : H \rightarrow H$ and consider $dX(t) = (AX(t) + F(X(t)))dt + BdW(t)$. The purely linear case above is the infinite-dimensional analogue of the Ornstein–Uhlenbeck process [3], and is the form we simulate.

This equation cannot, in general, be read as the integral equation

$$X(t) = x_0 + \int_0^t AX(s)ds + \int_0^t BdW(s)$$

Since the generator A is typically unbounded, there is no reason for the solution $X(s)$ to lie in the domain $D(A)$, so the integral $\int_0^t AX(s)ds$ may be undefined. To obtain a workable notion of solution, we apply the method of variation of constants to the deterministic part and treat $BdW(t)$ as a forcing term, leading to the following.

Definition 10.3. An H -valued process $\{X(t)\}_{t \geq 0}$ is a *mild solution* of the stochastic evolution equation if it satisfies

$$X(t) = S_t x_0 + \int_0^t S_{t-s} BdW(s)$$

The term $Z(t) := \int_0^t S_{t-s} BdW(s)$ is called the *stochastic convolution*.

The stochastic integral above is understood coordinate-wise: writing $W(t) = \sum_n \beta_n(t) e_n$ for independent real-valued Wiener processes β_n and an orthonormal basis $\{e_n\}$ of H , each projection $\langle Z(t), e_n \rangle$ is an ordinary real-valued Itô integral. The general construction of the H -valued stochastic integral can be found in [2]; we will only need the explicit diagonal form appearing in the heat equation example below.

Remark 10.4. The notion of a mild solution has several features worth noting.

1. It never requires $X(s) \in D(A)$, and may exist even when no classical (strong) solution does.
2. A mild solution is in general *not* an Itô process, so Itô's formula must be applied with care in infinite dimensions.
3. The stochastic convolution $Z(t)$ is a centered Gaussian with covariance operator

$$\int_0^t S_{t-s} B B^* S_{t-s}^* ds$$

and defines a genuine H -valued random variable precisely when this operator is trace-class.

11 Numerical Method

A natural choice for stochastic evolution is over a semigroup which is exponential decay over each basis vector. We will use the following equation as our case study.

11.1 Heat Equation

The subject of the following simulations is the heat equation, which, as described above, can be written as

$$\partial u_t = \partial u_{xx} = \Delta u$$

Similarly to [Example 9.4](#), given the Hilbert space $L^2([0, 1])$ and the natural sine basis, we can describe an initial condition of the heat equation u_0 as

$$u_0(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \approx \sum_{n=1}^k a_n \sin(n\pi x)$$

for a sufficiently large $k > 1$. As the basis vectors $\sin(n\pi x)$ are eigenvalues of the operator Δ with eigenvalue $-(n\pi)^2$, we have that

$$\frac{du_k}{dt} = -(k\pi)^2 u_k$$

which gives an exact solution

$$u_k(x, t) = a_k e^{-(k\pi)^2 t} \sin(k\pi x)$$

for the projection onto $\sin(k\pi x)$. Then

$$u(x, t) = \sum_{n=1}^k u_n(x, t) = \sum_{n=1}^k a_n e^{-(n\pi)^2 t} \sin(n\pi x)$$

We can also view the decay of the Fourier coefficients a_k as

$$a_{k,t} = a_k e^{-(k\pi)^2 t}$$

And in differential form as

$$da_k = -(k\pi)^2 a_k dt$$

Then considering all the coefficients together as a vector A_t , we have that

$$dA_{t,k} = -(k\pi)^2 A_{t,k} dt \quad (1)$$

Now that we have a deterministic solution over the infinite dimensional space, we can introduce the cylindrical noise. For the purposes of simulation, we will consider the cylindrical sets which are Borel sets on the first k Fourier coefficients, for a chosen k .

11.2 Stochastic Heat Equation

Let W be a Q -Wiener process over $L^2(0, 1)$ with the sine basis $e_k = \sin(k\pi x)$. We can add the stochastic term to (1):

$$dA_{t,k} = -(k\pi)^2 A_{t,k} dt + dW_k(t)$$

This is a stochastic evolution equation over the semigroup $\{S_t\}$, where

$$S_t e_k = e^{-(k\pi)^2 t} e_k$$

The mild solution can be written as

$$A_t = S_t A_0 + \int_0^t S_{t-s} dW(s)$$

11.2.1 Exact Solution

Using property 3 in [Remark 10.4](#), we can find the exact distribution of the integral

$$I_{t_1, t_2} = \int_{t_1}^{t_2} S_{t-s} dW(s) \sim \mathcal{N}\left(0, \int_{t_1}^{t_2} S_{t-s} S_{t-s}^* ds\right)$$

by solving for the covariance $Q_{t_1, t_2, k}$ over each basis vector, using the fact that S_t is self-adjoint:

$$Q_{t_1, t_2, k} = \int_{t_1}^{t_2} e^{-2(k\pi)^2(t-s)} ds = \frac{1}{2(k\pi)^2} \left(1 - e^{-2(k\pi)^2(t_2-t_1)}\right)$$

so that

$$I_{t_1, t_2} \sim \mathcal{N}(0, Q_{t_1, t_2})$$

If we take $\Delta t = t_n - t_{n-1}$ and denote

$$I_{t_{n-1}, t_n} = I_{\Delta t} \sim \mathcal{N}(0, Q_{\Delta t})$$

We can use this fact to derive an recursive exact solution for the heat equation with increment Δt :

$$\begin{aligned} A_{t_n, k} &= e^{-(k\pi)^2 t_n} A_{0, k} + \int_0^{t_n} e^{-(k\pi)^2(t_n-s)} dW_k(s) \\ &= e^{-(k\pi)^2 t_n} A_{0, k} + \int_0^{t_{n-1}} e^{-(k\pi)^2[(t_{n-1}+\Delta t)-s]} dW_k(s) + \int_{t_{n-1}}^{t_n} e^{-(k\pi)^2(t_n-s)} dW_k(s) \\ &= e^{-(k\pi)^2 \Delta t} \left[e^{-(k\pi)^2 t_{n-1}} A_{0, k} + \int_0^{t_{n-1}} e^{-(k\pi)^2(t_{n-1}-s)} dW_k(s) \right] + I_{\Delta t, k} \\ &= e^{-(k\pi)^2 \Delta t} A_{t_{n-1}, k} + I_{\Delta t, k} \end{aligned}$$

Algorithm 1: Heat Equation Spectral Estimation

Input: $\mathbf{u}_0$, the function $[0, 1] \rightarrow \mathbb{R}$ specifying the initial condition
 T , the length of the simulation
 N_t the size of the discretized grid over time
 k , the number of sine transform coefficients with which we estimate the curve
 $\mathbf{Q}$, the covariance matrix of size $k \times k$
 σ , the variance scale factor for the noise

Output: An $N_t \times k$ matrix A where $A_{ij} = \hat{u}_i(j)$, the estimated value at time i and position j

```
1 Function  $dst(\mathbf{u}, N)$ 
2    $\mathbf{a}_k \leftarrow \sum_{n=0}^{N-1} \mathbf{u}_n \sin[\frac{\pi}{N+1}(n+1)(k+1)]$ 
3   return  $\mathbf{a}$ 
4 end

5 Function  $idst(\mathbf{u}, N)$ 
6   return  $\frac{2}{N+1} dst(\mathbf{u}, N)$ 
7 end

8  $\mathbf{a} \leftarrow dst(\mathbf{u}_0, k)$ 
9  $A_{0j} \leftarrow u_0(j/k)$ 
10  $\Delta t \leftarrow T/N_t$ 
11 for  $t \in (0, \Delta t, \dots, T)$ 
12    $\mathbf{w} \leftarrow \mathcal{N}(0, \Delta t \mathbf{Q})$ 
13   for  $a_k \in \mathbf{a}$ 
14      $a_k \leftarrow e^{-(n\pi)^2 \Delta t} (a_k + \sigma \mathbf{w}_k)$ 
15    $A_k \leftarrow idst(a_k, k)$ 
16 return  $A$ 
```

11.2.2 Euler-Maruyama Estimate

Alternatively, instead of finding an exact distribution for the Itô integral, we can approximate it given small Δt as a rectangle with width $W_k(\Delta t)$ and height $e^{-(k\pi)^2\Delta t}$, giving the new recursion

$$A_{t_n,k} \approx e^{-(k\pi)^2\Delta t} [A_{t_{n-1},k} + W_k(\Delta t)]$$

With this method of recursive simulation, we provide the following algorithm.

11.3 Simulation Results

We now apply the spectral estimation scheme of [Algorithm 1](#) to the stochastic heat equation on $(0, 1)$. To explore the effect of the two main parameters — the initial condition u_0 and the covariance operator Q — we run the simulation for two initial conditions and three covariances, over the time interval $[0, T]$ with $T = 5$, using $k = 50$ sine modes. The first initial condition, $u_0^{(a)}(x) = 5x(x - 1)\sin(23\pi x)$, is an extremely bumpy profile, while the second, $u_0^{(b)}(x) = 4x(1 - x)$, is a single smooth hump; both are 0 at the endpoints, as required by the boundary conditions. The three covariances are all diagonal and differ only in how the noise intensity Q_n is distributed across the frequencies: $Q_1 = I$ assigns equal weight to every sine component (the cylindrical case); Q_2 has entries $Q_n = 10/n^2$; and Q_3 has entries $Q_n = 10e^{-n/4}$. The six resulting sample paths are shown in [Figure 1](#).

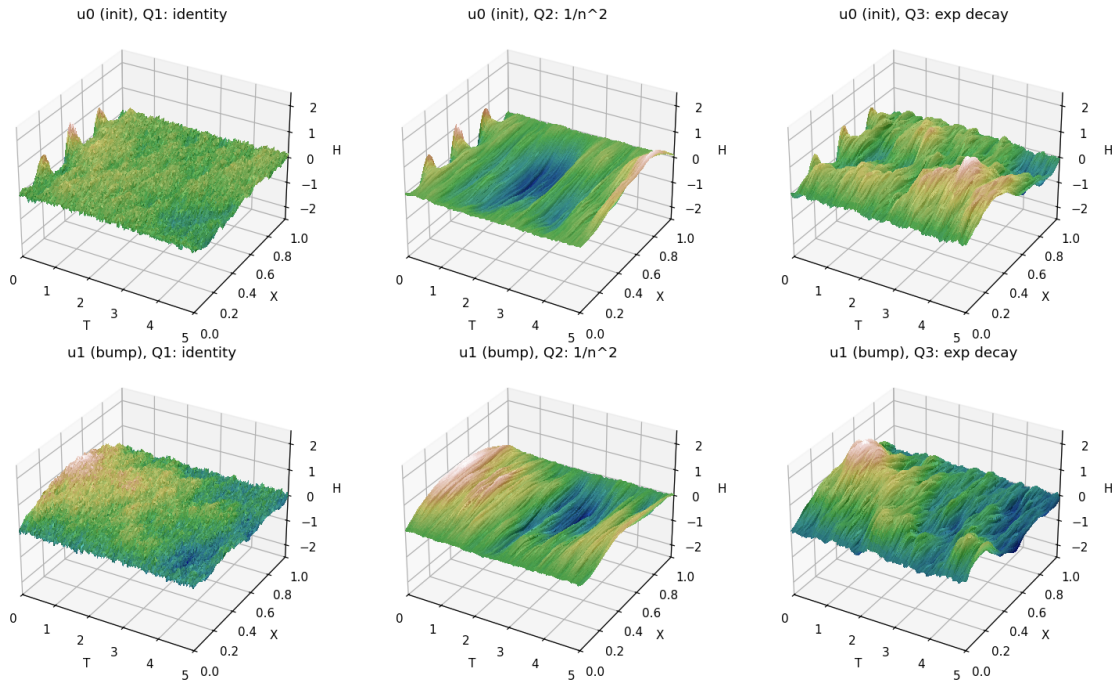


Figure 1 Single realisations of the stochastic heat equation over $[0, T]$ with $T = 5$, for two initial conditions (rows) and three covariance operators (columns). The vertical axis H is the temperature, with time T and space X on the horizontal axes.

The effect of the covariance. Comparing the three columns, the choice of Q controls the texture of the noise. Under $Q_1 = I$ every sine component, including the most “bumpy”, is excited equally, so the surface underlying curve is covered with lots of spikes with a little amplitude. For Q_2 and Q_3 where more bumpy components decay quicker the high frequencies die off and the randomness is visible in the smooth, low-frequency modes. The result is visibly different: Q_2 and Q_3 produce smoother curves.

The effect of the initial condition. The two rows show that the initial condition is clearly linked the early-time appearance but that effect goes steadily. The oscillatory profile $u_0^{(a)}$ carries most of its energy in high modes, which decay at the rapid rate $(n\pi)^2$; its structure is therefore smoothed away almost immediately, leaving only a couple of transient features near $t = 0$ before the dynamics become noise-dominated. The smooth hump $u_0^{(b)}$, by contrast, lives almost entirely in the slowest mode and so persists far longer, remaining recognisable well into the run. At later times, however, the influence of u_0 fades in both cases and each surface is dominated by the same Q -driven fluctuations: the texture depends on the noise, not on where the system started.

Single realisations versus the distribution. It is important to stress that each panel of [Figure 1](#) is a *single* sample path, obtained from one particular realisation of the driving noise. Unlike the deterministic heat equation, whose solution is a single determined surface, the stochastic equation has no single solution: rerunning with a different random seed produces a different-looking surface, and no individual path settles down to a fixed shape, since fresh noise is injected at every step. These plots should therefore be read as illustrations of the *texture* produced by each (u_0, Q) pair, not as the solution itself. We turn next to looking at the distribution of solutions the stochastic differential equation produces.

11.4 The Distribution of the Solution

The plots in the previous section each show a single sample path and therefore cannot describe the random solution on their own. To characterise the distribution of the solution we instead estimate its statistics directly: we fix an initial condition and covariance, draw $N = 300$ independent realisations, and for a given t compute the sample mean and the sample standard deviation across the curve. Both quantities are estimated $x_j = j/N_x$, $j = 0, \dots, N_x$, with $N_x = 50$ (so $N_x + 1 = 51$ points spanning $[0, 1]$ including the endpoints):

$$\bar{u}(x_j) = \frac{1}{N} \sum_{i=1}^N u_i(x_j), \quad S(x_j) = \left(\frac{1}{N-1} \sum_{i=1}^N (u_i(x_j) - \bar{u}(x_j))^2 \right)^{1/2}, \quad j = 0, \dots, N_x.$$

Then these points can be used to create an estimate. Since the equation is linear and the driving noise is Gaussian, the solution at any fixed time is itself Gaussian, so the mean and standard deviation determine the distribution completely; the $\bar{u} \pm S$ and $\bar{u} \pm 2S$ envelopes then contain roughly 68% and 95% of the realisations at each point.

Two opposing trends are visible between the panels. The mean decays: its peak falls from roughly 0.69 at $t = 2$ to about 0.40 at $t = 5$. This is exactly the deterministic behaviour, since taking the expectation of the evolution equation annihilates the mean-zero noise and leaves $\mathbb{E}[A_t] = S_t A_0$; the mean of the stochastic solution therefore satisfies the ordinary heat equation and relaxes toward the flat state. The hump persists as long as it does because u_0 is dominated by the slowest mode, which decays only slowly. At the same time the bands widen: the spread is larger at $t = 5$ than at $t = 2$, as the accumulated noise builds toward its stationary level. These are the two halves of the approach to equilibrium discussed earlier — the mean sinking toward zero while the variance rises toward a constant.

The shape of the spread is also informative. Both the mean and the standard deviation are pinned to zero at the endpoints $x = 0$ and $x = 1$: the Dirichlet boundary conditions fix the solution there in every realisation, so there is no room for fluctuation. The bands are therefore widest in the interior and pinch shut at the boundaries, giving the envelope its characteristic lens shape. Finally, the bands should be read as a summary of the *ensemble* rather than of any single trajectory: an individual realisation is a rough, jittery curve that wanders within these limits and

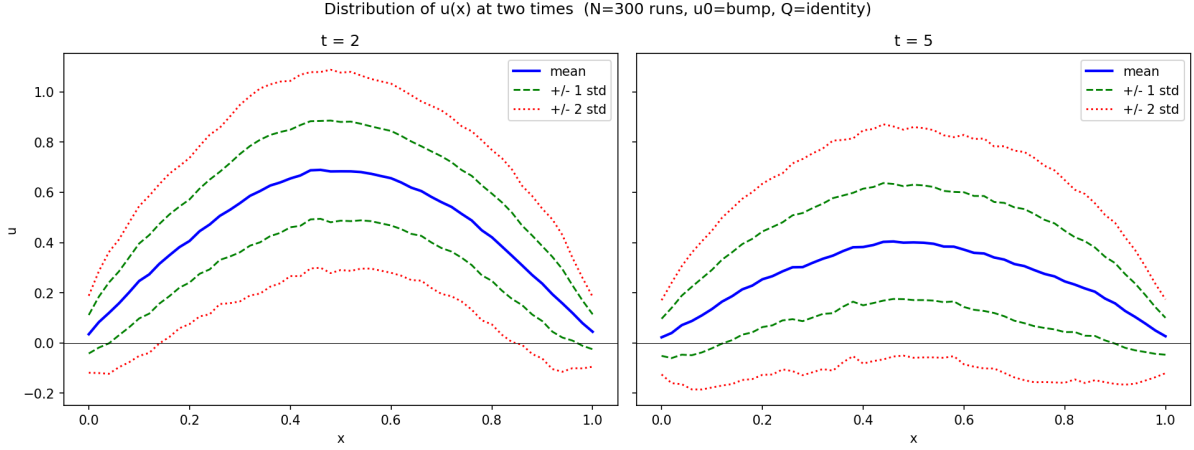


Figure 2 Estimated distribution of $u(x)$ at $t = 2$ (left) and $t = 5$ (right), from $N = 300$ realisations with $u_0(x) = 4x(1 - x)$ and $Q = I$. The solid curve is the sample mean and the dashed and dotted curves are the ± 1 and ± 2 standard-deviation bands.

never settles, whereas the mean and standard deviation are smooth, well-defined functions that fully describe the distribution from which those trajectories are drawn.

The lens shape above is a consequence of the covariance choice $Q = I$, which excites every sine component and so spreads the uncertainty smoothly across the interior. A more structured covariance reshapes the spread accordingly. [Figure 3](#) repeats the experiment with the sign-changing initial condition $u_0(x) = \sin(2\pi x)$ and a covariance that excites only the third mode, $Q = 10 e_3 e_3^\top$. The mean retains the two-lobe profile of u_0 and decays as before, but the standard deviation now inherits the shape of $\sin(3\pi x)$: the deviations pinch to zero not only at the boundaries but also at the interior nodes $x = 1/3$ and $x = 2/3$, where that mode vanishes. The contrast with [Figure 2](#) makes the underlying structure explicit — the shape of the mean is determined by the initial condition u_0 , while the shape of the spread is determined by the covariance Q , and the two need not coincide.

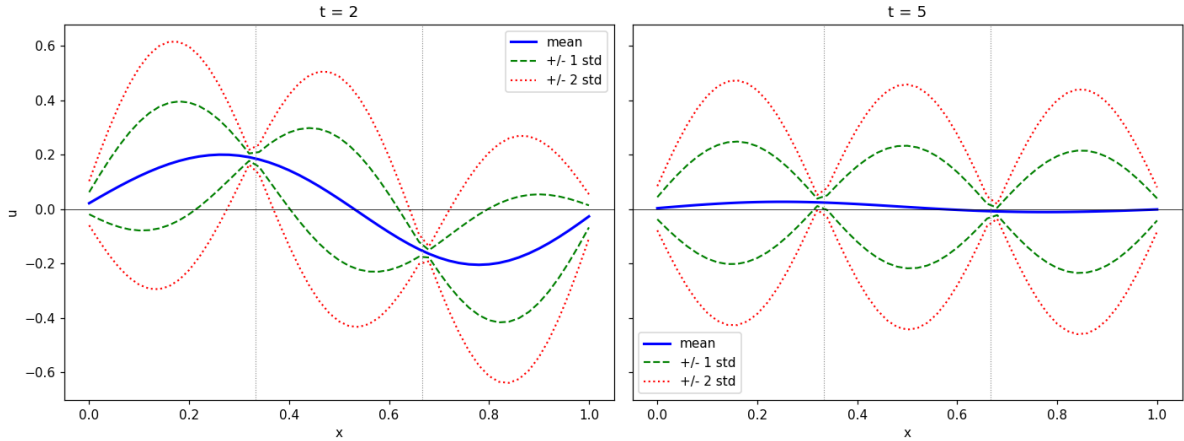


Figure 3 Estimated distribution of $u(x)$ at $t = 2$ (left) and $t = 5$ (right), from $N = 300$ realisations with $u_0(x) = \sin(2\pi x)$ and a covariance exciting only the third sine mode. The mean keeps the two-lobe shape of u_0 , while the standard-deviation bands vanish at the interior nodes $x = 1/3$ and $x = 2/3$ of $\sin(3\pi x)$ (dotted vertical lines).

12 Approximation vs Exact Solution

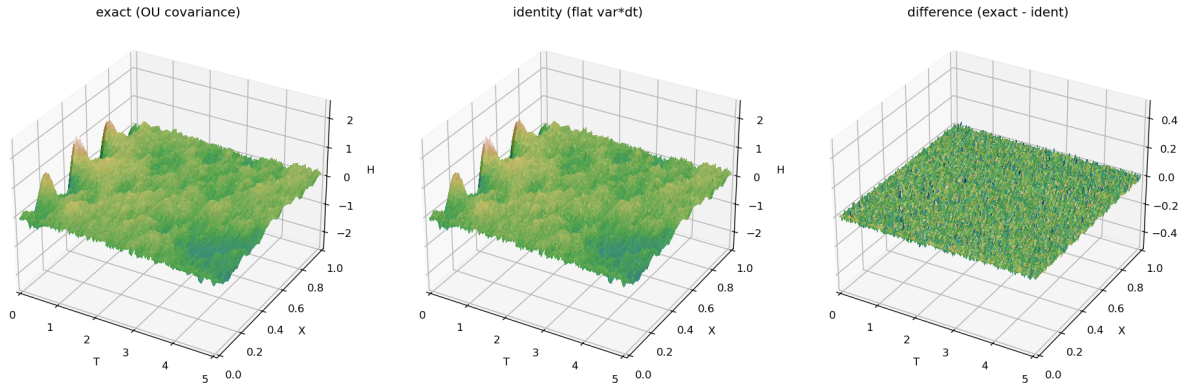


Figure 4 Solution under the exact per-mode noise covariance (left), the flat identity approximation (centre), and their difference (right, note the rescaled axis). Same seed throughout; the discrepancy stays below ≈ 0.05 .

We considered sampling using the exact OU increment covariance,

$$Q_{\Delta t, k} = \frac{1 - e^{-2\lambda_k \Delta t}}{2\lambda_k} = \Delta t + O(\Delta t^2), \quad \lambda_k = d(k\pi)^2$$

but our Δt is small enough that $e^{-2\lambda_k \Delta t} \approx 1$ for the energetic modes, so $Q_{\Delta t, k} \approx \Delta t$: the flat identity approximation. Figure 4 confirms the two are indistinguishable, so we keep the approximation.

13 Discussion and Future Work

- try different SDEs (non-linear case?) - try different error measurement

Statement on the Use of Generative AI

The group used generative AI tools in a limited supporting capacity during the preparation of this project. In particular, such tools were used to assist with:

- improving the organisation of the report and the clarity of exposition
- suggesting alternative wording for definitions, transition paragraphs, and summaries
- checking grammar and $\text{\LaTeX}$ formatting
- discussing possible approaches to numerical illustrations and presentation structure

Generative AI tools were not used as a substitute for the cited mathematical sources. All definitions, theorems, proofs, numerical methods, images, and code included in the final report were verified by the authors against the relevant references or by direct verification. The authors assume full responsibility for the accuracy, originality, referencing, and presentation of the final submitted work.

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